

Chapter 3

Geometric Class Field Theory

The purpose of this chapter is to deduce the reduced form of geometric class field theory, [Theorem 2](#), from the ramification calculation established in Chapter 2.

Let C be a smooth, projective, geometrically connected curve over the perfect field k , let $\mathfrak{m}$ be a modulus on C , and put $U = C \setminus \mathfrak{m}$. Given a rank-one local system $\mathcal{F}$ on U whose ramification is bounded by $\mathfrak{m}$, our goal is to descend its symmetric power $\mathcal{F}^{[d]}$ along the Abel–Jacobi morphism

$$\Phi_d: U^{(d)} \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}^d$$

for all sufficiently large d .

When $\mathfrak{m} = 0$, the fibers of the Abel–Jacobi morphism are projective spaces. Their trivial étale fundamental groups allow the desired descent directly. When $\mathfrak{m} > 0$, however, the fibers of Φ_d are affine spaces, so this argument cannot be applied to Φ_d itself. Following Takeuchi, we resolve this difficulty by extending Φ_d to a projective space bundle

$$\tilde{\Phi}_d: \tilde{C}_{\mathfrak{m}}^{(d)} \longrightarrow \mathrm{Pic}_{C,\mathfrak{m}}^d,$$

where $\tilde{C}_{\mathfrak{m}}^{(d)}$ is an open subscheme of the blowup of $C^{(d)}$ along the locus obtained by adding the modulus.

The proof then has three remaining steps. First, we use the single-point ramification calculation of Chapter 2, together with compatibility under products and addition of divisors, to prove that $\mathcal{F}^{[d]}$ is tamely ramified along the boundary

$$H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}.$$

Second, we show that in this full-modulus compactification the tame ramification is in fact unramified, and hence that $\mathcal{F}^{[d]}$ extends uniquely to a local system on $\tilde{C}_{\mathfrak{m}}^{(d)}$. Finally, since $\tilde{\Phi}_d$ is a projective space bundle, it induces an isomorphism on étale fundamental groups. The extended local system therefore descends uniquely to $\mathrm{Pic}_{C,\mathfrak{m}}^d$, proving [Theorem 2](#).

We begin by recalling the generalized Picard scheme and the geometry of the Abel–Jacobi morphism. We then introduce Takeuchi’s compactification, prove the required ramification and extension results, and conclude with the descent argument.

3.1 Generalized Picard Schemes and the Abel–Jacobi Map

3.1.1 The generalized Picard scheme

We recall the notion of generalized Jacobian varieties and study their fundamental properties. The material presented here is primarily adapted from [Gui19] and [Tak19]. For further background on the general theory of abelian varieties and Jacobians, the reader may also consult [Mil08]. Let S be a scheme and let C be a projective smooth S -scheme whose geometric fibers are connected and of dimension 1. Let $\mathfrak{m}$ be a modulus on C , defined as an effective Cartier divisor of C/S (i.e., a closed subscheme of C which is finite flat of finite presentation over S). We denote the projection $C \times_S T \rightarrow T$ by pr for any S -scheme T .

The Functor of Points

Let d be an integer. For an S -scheme T , we consider the set of data $(\mathcal{L}, \psi)$ where:

- $\mathcal{L}$ is an invertible sheaf of degree d on C_T .
- $\psi : \mathcal{O}_{\mathfrak{m}_T} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}_T}$ is a trivialization of $\mathcal{L}$ along the modulus.

Two such pairs $(\mathcal{L}, \psi)$ and $(\mathcal{L}', \psi')$ are said to be isomorphic if there exists an isomorphism of invertible sheaves $f : \mathcal{L} \rightarrow \mathcal{L}'$ such that the following diagram commutes:

$$\begin{array}{ccc} & \mathcal{O}_{\mathfrak{m}_T} & \\ \psi' \swarrow & & \searrow \psi \\ \mathcal{L}'|_{\mathfrak{m}_T} & \xrightarrow{f|_{\mathfrak{m}_T}} & \mathcal{L}|_{\mathfrak{m}_T} \end{array}$$

We define the presheaf $\text{Pic}_{C,\mathfrak{m}}^{d,\text{pre}}$ on Sch/S by assigning to T the set of isomorphism classes of such pairs. Let $\text{Pic}_{C,\mathfrak{m}}^d$ denote the étale sheafification of this presheaf.

Representability and Structure

The fundamental properties of this functor are as follows:

1. $\text{Pic}_{C,\mathfrak{m}}^d$ is represented by an S -scheme. (Note: If $\mathfrak{m}$ is faithfully flat over S , the presheaf is already a étale sheaf).
2. $\text{Pic}_{C,\mathfrak{m}}^0$ is a smooth commutative group S -scheme with geometrically connected fibers, referred to as the *generalized Jacobian variety* of C with modulus $\mathfrak{m}$.
3. For any d , $\text{Pic}_{C,\mathfrak{m}}^d$ is a $\text{Pic}_{C,\mathfrak{m}}^0$ -torsor.

In the case where $\mathfrak{m} = 0$, we recover the standard Jacobian variety, denoted simply as Pic_C^d .

Relation to the Standard Jacobian

We now examine the behavior of the generalized Picard scheme under the variation of the modulus. By viewing the structure along the modulus as an additional rigidification, we obtain natural transition maps corresponding to the inclusion of moduli.

Let $\mathfrak{m}_1$ and $\mathfrak{m}_2$ be moduli such that $\mathfrak{m}_1 \subset \mathfrak{m}_2$. There exists a natural map

$$\mathrm{Pic}_{C, \mathfrak{m}_2}^d \rightarrow \mathrm{Pic}_{C, \mathfrak{m}_1}^d$$

obtained by restricting the isomorphism ψ . Since $\mathfrak{m}_2$ is a finite S -scheme, this map is a surjection as a morphism of étale sheaves. In particular, for any modulus $\mathfrak{m}$, there is a natural surjective morphism of étale sheaves:

$$\mathrm{Pic}_{C, \mathfrak{m}}^d \rightarrow \mathrm{Pic}_C^d.$$

Local Freeness and Base Change

Let $\mathfrak{m}$ be a modulus which is everywhere strictly positive. Let g denote the genus of C , which is a locally constant function on S . We restrict our attention to degrees d satisfying the condition:

$$d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}. \quad (3.1)$$

Assuming S is quasi-compact, such a d always exists.

Fix an integer d satisfying the condition above. Let T be an S -scheme and let $\mathcal{L}$ be an invertible sheaf of degree d on C_T . One can show that the pushforwards $\mathrm{pr}_* \mathcal{L}$ and $\mathrm{pr}_* \mathcal{L}(-\mathfrak{m})$ are locally free sheaves and their formations commute with any base change. Explicitly, for any morphism of S -schemes $f : T' \rightarrow T$, the base change morphisms are isomorphisms:

$$f^* \mathrm{pr}_* \mathcal{L} \xrightarrow{\sim} \mathrm{pr}_* f^* \mathcal{L}$$

and

$$f^* \mathrm{pr}_* (\mathcal{L}(-\mathfrak{m})) \xrightarrow{\sim} \mathrm{pr}_* f^* (\mathcal{L}(-\mathfrak{m})).$$

In particular, following [Gui19], if $\mathcal{L}$ is invertible $\mathcal{O}_C$ -module with degree d satisfying 3.1 on each fiber of f then, $\mathrm{pr}_* \mathcal{L}$ is a locally free $\mathcal{O}_S$ -module of rank $d - g + 1$.

For further background and verification of these constructions, we refer the reader to Milne's notes on abelian Varieties ([Mil08]).

3.1.2 The Abel–Jacobi Morphism and Its Fibers

Let $U = C \setminus \mathfrak{m}$ be the complement of the modulus in C . The effective cartier divisors of degree d which are prime to $\mathfrak{m}$ are parameterized by the symmetric power $\mathrm{Sym}_S^d(U) = U^{(d)}$ over S (See [Gui19] Proposition 4.12, [Mil08] Theorem 3.13). For any such divisor $D \in U^{(d)}$, the associated line bundle $\mathcal{O}_C(D)$ admits a canonical trivialization along $\mathfrak{m}$. Specifically, the canonical section 1_D is regular and non-vanishing on $\mathfrak{m}$ because $\mathrm{supp}(D) \cap \mathrm{supp}(\mathfrak{m}) = \emptyset$. This section restricts to a nowhere-vanishing section on the subscheme $\mathfrak{m}$, thereby determining a trivialization $\psi_D^{-1} : \mathcal{O}_C(D)|_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{O}_{\mathfrak{m}}$. This is done functorially in families, yielding a morphism from the symmetric power to the generalized Picard scheme (over S):

$$\Phi_d : U^{(d)} \rightarrow \mathrm{Pic}_{C, \mathfrak{m}}^d, \quad D \mapsto [(\mathcal{O}_C(D), \psi_D)], \quad (3.2)$$

When $\mathbf{m} = 0$, $d \geq \max\{2g-1, 0\}$ and C admits a section over S , $C^{(d)}$ is a projective space bundle over Pic_C^d . It is proper, surjective with geometrically connected fibers.

Guignard ([Gui19] Theorem 4.14) proves that for $\mathbf{m} > 0$ and d satisfying (3.1), the Abel-Jacobi morphism Φ_d is surjective smooth of relative dimension $d - \deg \mathbf{m} - g + 1$, with geometrically connected fibers.

When $S = \text{Spec}(k)$, the geometric-fibers of Φ_d are well understood:

Theorem 59. *Assuming $S = \text{Spec}(k)$ and $d \geq \max\{2g-1 + \deg \mathbf{m}, \deg \mathbf{m}\}$. Then, the geometric-fibers of the Abel-Jacobi morphism*

$$\Phi_d : U^{(d)} \rightarrow \text{Pic}_{C,\mathbf{m}}^d$$

over any point are isomorphic to

$$\begin{cases} \mathbb{A}_{k^{sep}}^{d-\deg \mathbf{m}-g+1} & \text{if } \mathbf{m} > 0 \\ \mathbb{P}_{k^{sep}}^{d-g} & \text{if } \mathbf{m} = 0 \end{cases}$$

In both cases Φ_d is a fibration in affine spaces or projective spaces, depending on whether $\mathbf{m}$ is non-zero or zero.

Proof. See [Ten15] Propositions 3.13-3.14, or [T6t11] Proposition 2.1.4. □

3.2 Takeuchi's Compactification of the Abel–Jacobi Morphism

We recall that our objective is to descend the local system $\mathcal{F}^{[d]}$ from $U^{(d)}$ to $\text{Pic}_{C,\mathbf{m}}^d$ along the Abel-Jacobi map Φ_d :

$$\begin{array}{ccc} \mathcal{F}^{[d]} & & \\ \downarrow & & \\ U^{(d)} & \xrightarrow{\Phi_d} & \text{Pic}_{C,\mathbf{m}}^d \end{array}$$

(Here, the purple arrow emphasizes that the morphism is of sheaves on the étale site).

However, we encounter an obstruction: in the case we are considering ($\mathbf{m} > 0$), the fibers of Φ_d are affine spaces (of the same degree) rather than the better-behaved projective spaces. This suggests that a solution to this problem is to compactify the morphism to yield projective fibers.

This section describes the result of the compactification constructed by [Tak19] by blowing up the symmetric power.

Let $\mathbf{m} = \sum_{i=1}^n k_i P$ with $\deg P = d_P$ be a modulus on C , and let d satisfy (3.1). Takeuchi ([Tak19]) defines $Z_0 = Z_0(\mathbf{m}, d)$ as the closed subscheme of $C^{(d)}$ defined by the map $C^{(d-\deg \mathbf{m})} \rightarrow C^{(d)}$ adding $\mathbf{m}$. He also defines $X_{\mathbf{m},d}$ as the blowup of $C^{(d)}$ along Z_0 . Let $E_0 = E_{\mathbf{m},d} = Z_0(\mathbf{m}, d) \times_{C^{(d)}} X_{\mathbf{m},d}$ be the exceptional divisor of the blowup. It is irreducible of codimension 1, and we let $\eta_0 = \eta_{\mathbf{m},d}$ be its generic point.

Diagrammatically:

$$\begin{array}{ccc} \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} \\ \downarrow & & \downarrow \pi \\ Z_0 & \xrightarrow{c.i} & C^{(d)} \end{array}$$

Incorporating $U^{(d)}$, the local system $\mathcal{F}^{[d]}$ and the Abel-Jacobi map, we have:

$$\begin{array}{ccccccc} & & & \mathcal{F}^{[d]} & & & \\ & & & \downarrow & & & \\ \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} & \longleftrightarrow & U^{(d)} & \xrightarrow{\Phi_d} & \text{Pic}_{C,\mathfrak{m}}^d \\ \downarrow & & \downarrow \pi & \swarrow & \downarrow & & \\ Z_0 & \xrightarrow{c.i} & C^{(d)} & & & & \end{array}$$

In Section 3 of [Tak19] Takeuchi constructs, for d satisfying (3.1) a compactification denoted by $\tilde{C}_{\mathfrak{m}}^{(d)}$ and proves the following:

Theorem 60 (Takeuchi). *The scheme $\tilde{C}_{\mathfrak{m}}^{(d)}$ is an open subscheme of $X_{\mathfrak{m},d}$ containing $U^{(d)}$. The morphism $\Phi_d : U^{(d)} \rightarrow \text{Pic}_{C,\mathfrak{m}}^d$ extends to a morphism $\tilde{\Phi}_d : \tilde{C}_{\mathfrak{m}}^{(d)} \rightarrow \text{Pic}_{C,\mathfrak{m}}^d$ which makes $\tilde{C}_{\mathfrak{m}}^{(d)}$ a projective space bundle over $\text{Pic}_{C,\mathfrak{m}}^d$. Furthermore, the complement of $U^{(d)}$ in $\tilde{C}_{\mathfrak{m}}^{(d)}$ is the scheme-theoretic intersection*

$$E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}_{\mathfrak{m}}^{(d)} \cong Z_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}.$$

Diagrammatically we have:

$$\begin{array}{ccccccc} & & & \mathcal{F}^{[d]} & & & \\ & & & \downarrow & & & \\ E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}_{\mathfrak{m}}^{(d)} & \longrightarrow & \tilde{C}_{\mathfrak{m}}^{(d)} & \longleftrightarrow & U^{(d)} & & \\ \downarrow & & \downarrow & \searrow \tilde{\Phi}_d & \downarrow \Phi_d & & \\ \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} & & \text{Pic}_{C,\mathfrak{m}}^d & & \\ \downarrow & & \downarrow \pi & & & & \\ Z_0 & \xrightarrow{c.i} & C^{(d)} & & & & \end{array}$$

Here the boundary may equivalently be written as $Z_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$, since $E_0 = Z_0 \times_{C^{(d)}} X_{\mathfrak{m},d}$.

3.3 Tame Ramification Along the Full Boundary

The ramification calculation of Chapter 2 concerns a modulus supported at a single point and a symmetric power of the corresponding degree. We now combine those local calculations over the different points of $\mathfrak{m}$ and then pass from degree $\deg \mathfrak{m}$ to every sufficiently large degree d .

Theorem 61. *Let Λ be a finite ring of cardinality invertible in k , and let $\mathcal{F}$ be an étale sheaf of Λ -modules, locally free of rank 1 on U , with ramification bounded by $\mathfrak{m}$. For every sufficiently large integer d , the local system $\mathcal{F}^{[d]}$ is tamely ramified along*

$$H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)} = E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}_{\mathfrak{m}}^{(d)} \cong Z_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}.$$

3.3.1 Compatibility Under Products

We first record the product-compatibility result needed to combine the point-supported ramification calculations of Chapter 2.

Let X and Y be smooth schemes over a field k , and let $x \in X$ and $y \in Y$ be closed points. We denote the blowups of these schemes at the given points by $\pi_X : \text{Bl}_x(X) \rightarrow X$ and $\pi_Y : \text{Bl}_y(Y) \rightarrow Y$. Furthermore, let $\pi_{X \times Y} : \text{Bl}_{(x,y)}(X \times_k Y) \rightarrow X \times_k Y$ be the blowup of the product scheme at the point (x, y) . We denote by E_X, E_Y , and $E_{X \times Y}$ the respective exceptional divisors, and let η_X, η_Y , and $\eta_{X \times Y}$ be their generic points.

In this section, we establish the following result concerning the stability of ramification bounds under the external product of torsors.

Proposition 62. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G -torsors defined on open subsets $U_X \subset \text{Bl}_x(X)$ and $U_Y \subset \text{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If the ramification of $\mathcal{G}_X$ at η_X and $\mathcal{G}_Y$ at η_Y is bounded by r , then the external product torsors*

$$\mathcal{G}_{X \times Y} := pr_1^{-1} \mathcal{G}_X \otimes pr_2^{-1} \mathcal{G}_Y$$

have ramification bounded by r at the generic point $\eta_{X \times Y}$ of the exceptional divisor in the product blowup.

Proof. The proposition is purely local in nature, so it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y , we may restrict our attention to open neighborhoods of x and y that are étale over affine spaces. The rest of this section treats that case.

The Affine Case

Let $X = \mathbb{A}_k^n$ be the affine n -space over a field k , and let $0 \in X$ be the origin. Let $\tilde{X} = \text{Bl}_0(X)$ be the blowup of X at the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}_k^{n-1}$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}_k^{n-1}$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E , and let $R = \mathcal{O}_{\tilde{X}, \eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $\tilde{K} = K(X) = k(x_1, \dots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1} x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k \left[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1} \right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \dots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R .

For any monomial $M = x_1^{a_1} \dots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \dots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R , the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \dots + f_l$, where f_i is the homogeneous part of degree i , we have $\nu_E(f) = d$ (the order of vanishing at the origin).

The Product Case

Now, let $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$ with origins $x = 0$ and $y = 0$. As before, $\text{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\text{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}_k^{n+m}$. The blowup of the product at the origin $(0,0)$, denoted $\text{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \leq i, j \leq n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \leq k, l \leq m \\ x_i w_{n+j} = y_j w_i & 1 \leq i \leq n, 1 \leq j \leq m \end{cases}$$

where $[w_1 : \dots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X \times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\text{Bl}_0(X) \times_k \text{Bl}_0(Y)$ and $\text{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$; their common dense open is

$$(X \setminus \{0\}) \times_k (Y \setminus \{0\}).$$

Put $B = \text{Bl}_{(0,0)}(X \times_k Y)$, and let

$$q: B \longrightarrow \mathbb{P}_k^{n+m-1}$$

be the second projection in the above realization of B . Define

$$\Omega_X = \bigcup_{i=1}^n D_+(w_i), \quad \Omega_Y = \bigcup_{j=1}^m D_+(w_{n+j}),$$

and

$$\tilde{U} := q^{-1}(\Omega_X \cap \Omega_Y) = \bigcup_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} q^{-1}(D_+(w_i w_{n+j})).$$

There is a morphism

$$\begin{aligned} f: \tilde{U} &\longrightarrow \text{Bl}_0(X) \times_k \text{Bl}_0(Y), \\ ((x, y), [w_1 : \dots : w_{n+m}]) &\longmapsto ((x, [w_1 : \dots : w_n]), (y, [w_{n+1} : \dots : w_{n+m}])). \end{aligned}$$

Thus we have a diagram

$$\begin{array}{ccc} & \tilde{U} & \\ f \swarrow & & \searrow j \\ \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & B, \end{array}$$

where j is the open immersion. The restriction of q induces an isomorphism $E_{X \times Y} \xrightarrow{\sim} \mathbb{P}_k^{n+m-1}$. Since neither $V_+(w_1, \dots, w_n)$ nor $V_+(w_{n+1}, \dots, w_{n+m})$ contains the generic point of $\mathbb{P}_k^{n+m-1}$, we have $\eta_{X \times Y} \in \tilde{U}$.

Extensions of DVRs

Let

$$S = \mathcal{O}_{B, \eta_{X \times Y}}, \quad R_X = \mathcal{O}_{\mathrm{Bl}_0(X), \eta_X}, \quad R_Y = \mathcal{O}_{\mathrm{Bl}_0(Y), \eta_Y}.$$

The morphisms $\mathrm{pr}_X \circ f$ and $\mathrm{pr}_Y \circ f$ send $\eta_{X \times Y}$ to η_X and η_Y , respectively, and therefore induce local homomorphisms of DVRs

$$R_X \longrightarrow S, \quad R_Y \longrightarrow S.$$

The point $\eta_{X \times Y}$ belongs to $q^{-1}(D_+(w_1 w_{n+1}))$. On this chart,

$$\begin{aligned} S &= k \left[x_1, \frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \dots, \frac{y_m}{x_1} \right]_{(x_1)} \\ &= k \left[y_1, \frac{x_1}{y_1}, \dots, \frac{x_n}{y_1}, \frac{y_2}{y_1}, \dots, \frac{y_m}{y_1} \right]_{(y_1)}. \end{aligned}$$

Consequently, x_1 is a uniformizer in both R_X and S , while y_1 is a uniformizer in both R_Y and S . Both extensions have ramification index one. Their residue-field extensions are

$$k \left(\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1} \right) \subset k \left(\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \dots, \frac{y_m}{x_1} \right)$$

and

$$k \left(\frac{y_2}{y_1}, \dots, \frac{y_m}{y_1} \right) \subset k \left(\frac{x_1}{y_1}, \dots, \frac{x_n}{y_1}, \frac{y_2}{y_1}, \dots, \frac{y_m}{y_1} \right),$$

respectively. In particular, both are purely transcendental and hence separable.

Ramification of G -Torsors

The extensions $R_X \rightarrow S$ and $R_Y \rightarrow S$ are weakly unramified and have separable residue-field extensions. Hence the pullbacks of $\mathcal{G}_X$ and $\mathcal{G}_Y$ to $\mathrm{Frac}(S)$ both have ramification bounded by r . Their contracted product is the restriction of $\mathcal{G}_{X \times Y}$ to $\mathrm{Frac}(S)$, so [Lemma 42](#) gives the same bound for $\mathcal{G}_{X \times Y}$ at $\eta_{X \times Y}$. $\square$

3.3.2 From Point-Supported Moduli to the Full Boundary

The following addition lemma is adapted from [\[Tak19, Lemma 4.1\]](#).

Lemma 63. *Let C be a projective, smooth, and geometrically connected curve over a perfect field k . Let $\mathfrak{m} = \sum_{i=1}^r k_i P_i$ be an effective divisor where $P_1, \dots, P_r$ are distinct closed points. Let $U = C \setminus \mathfrak{m}$ and let $d \geq \deg \mathfrak{m}$.*

Suppose $\mathfrak{n}_1, \dots, \mathfrak{n}_l$ are pairwise coprime submoduli of $\mathfrak{m}$ such that $\mathfrak{m} = \sum_{j=1}^l \mathfrak{n}_j$. Consider the summation morphism:

$$\pi : C^{(\deg \mathfrak{n}_1)} \times_k \dots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d - \deg \mathfrak{m})} \longrightarrow C^{(d)}$$

defined by $(D_1, \dots, D_l, D_{extra}) \mapsto \sum_{j=1}^l D_j + D_{extra}$.

Then π is étale at the generic point of the closed subvariety

$$V = \{\mathfrak{n}_1\} \times_k \dots \times_k \{\mathfrak{n}_l\} \times_k C^{(d - \deg \mathfrak{m})}$$

inside the domain $C^{(\deg \mathfrak{n}_1)} \times_k \dots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d - \deg \mathfrak{m})}$.

Proof. We may assume that k is algebraically closed (hence $\deg P_i = 1$ for all i). The morphism π is finite: it is induced by the finite quotient morphism

$$C^d / (\mathfrak{S}_{\deg \mathfrak{n}_1} \times \dots \times \mathfrak{S}_{\deg \mathfrak{n}_l} \times \mathfrak{S}_{d - \deg \mathfrak{m}}) \longrightarrow C^d / \mathfrak{S}_d.$$

Its source and target are smooth k -schemes of pure dimension d . Hence π is flat by miracle flatness, and therefore finite flat. It is enough to show that there exists a closed point Q of $\mathfrak{n}_1 + \dots + \mathfrak{n}_l + C^{(d - \deg \mathfrak{m})} \subset C^{(d)}$ over which there are $\deg \pi$ points on $C^{(\deg \mathfrak{n}_1)} \times_k \dots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d - \deg \mathfrak{m})}$. (Because it will be unramified at this point and thus also at the generic point of V .) Choose Q as a point corresponding to a divisor $\mathfrak{n}_1 + \dots + \mathfrak{n}_l + P_{r+1} + \dots + P_{r+d - \deg \mathfrak{m}}$, where $P_1, \dots, P_{r+d - \deg \mathfrak{m}}$ are distinct points of $U(k)$. $\square$

Corollary 64. *The morphism $C^{(\deg \mathfrak{m})} \times_k C^{(d - \deg \mathfrak{m})} \xrightarrow{\pi} C^{(d)}$ is finite flat everywhere, and étale at the generic point of the closed subvariety $Z_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d - \deg \mathfrak{m})}$.*

Proof. This is the case $l = 1$ and $\mathfrak{n}_1 = \mathfrak{m}$ of Lemma 63. $\square$

Following from this, we look at the following diagram, coming from the flat base change

$C^{(d - \deg \mathfrak{m})} \rightarrow \text{Spec}(k)$ (Proposition 20) of (2.1):

$$\begin{array}{ccccc} E_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})} & \longrightarrow & X_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})} & & \mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d - \deg \mathfrak{m})} \\ \downarrow & & \downarrow & \nwarrow & \downarrow \\ Z_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})} & \longrightarrow & C^{(\deg \mathfrak{m})} \times_k C^{(d - \deg \mathfrak{m})} & & U^{(\deg \mathfrak{m})} \times_k C^{(d - \deg \mathfrak{m})} \end{array}$$

Note that $U^{(\deg \mathfrak{m})} \times_k C^{(d - \deg \mathfrak{m})}$ is dense open subscheme of $X_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})}$, And $E_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})}$ is a prime divisor of $X_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})}$. Hence it is a well-defined question according to Definition 38 to ask whether $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d - \deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})}$.

Lemma 65. *If $\mathcal{F}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$, then $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d - \deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d - \deg \mathfrak{m})}$.*

Proof. This follows from [Stacks, Tag 0EYD] $\square$

By Lemma 65, replacing $C^{(d-\deg \mathfrak{m})}$ with the dense open subscheme $U^{(d-\deg \mathfrak{m})} \subset C^{(d-\deg \mathfrak{m})}$, we get that the G -torsor $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$ ($\mathcal{P}$ corresponds to $\mathcal{F}$ under Proposition 9) is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \subset U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$, where $p_1 : U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \rightarrow U^{(\deg \mathfrak{m})}$ is the projection to the first factor.

Looking at the second projection $p_2 : X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \rightarrow U^{(d-\deg \mathfrak{m})}$, and the fact that $\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $U^{(d-\deg \mathfrak{m})}$ we get that $p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]} = X_{\mathfrak{m}} \times_k \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Hence, its restriction to $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ is unramified at θ .

Thus, by Lemma 66, we conclude that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]} = p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$.

Lemma 66. *Let $X \rightarrow \text{Spec } k$ be a scheme over a field k , and let $\mathcal{P}_1, \mathcal{P}_2$ be two G -torsors on U_{et} . Let ξ be the generic point of a prime divisor $D \subset X$. If $\mathcal{P}_1$ is tamely ramified at ξ , and $\mathcal{P}_2$ is unramified at ξ , then the product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at ξ .*

Proof. This follows from Lemma 42. □

Combining this with Corollary 64 we get

Corollary 67. *Let $\mathfrak{m}$ be a modulus as above, and let $\eta_{\mathfrak{m}}$ be the generic point of $E_{\mathfrak{m}}$. Let $\mathcal{P}$ be a G -torsor on U_{et} with ramification bounded by $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Then $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$, and $\mathcal{P}^{[d]}$ is tamely ramified at the generic point $\eta_0 = \eta_{\mathfrak{m},d}$ of $E_0 = E_{\mathfrak{m},d}$.*

Proof. The first assertion, that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ , follows from the preceding discussion. Thus, it remains to show that $\mathcal{P}^{[d]}$ is tamely ramified at η_0 .

Consider the blowup diagram defining $X_{\mathfrak{m},d}$:

$$\begin{array}{ccc} \overline{\{\eta_0\}} = E_0 & \longrightarrow & X_{\mathfrak{m},d} \\ \downarrow & & \downarrow \pi \\ Z_0 & \xrightarrow{c.i} & C^{(d)} \end{array}$$

By performing a base change along the flat addition map $+: C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \rightarrow C^{(d)}$, we obtain the following commutative diagram:

$$\begin{array}{ccccc} \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \right) \times_{C^{(d)}} E_0 & \longrightarrow & \overline{\{\eta_0\}} = E_0 & & \\ \downarrow & & \downarrow & & \\ \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \right) \times_{C^{(d)}} X_{\mathfrak{m},d} & \longrightarrow & X_{\mathfrak{m},d} & & \\ \downarrow & & \downarrow \pi & & \\ \mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{c.i} & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{+} & C^{(d)} \end{array}$$

By Theorem 30, blowups commute with flat base change. Since the inverse image of the center Z_0 under the map $+$ is $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$, the scheme $\left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \right) \times_{C^{(d)}} X_{\mathfrak{m},d}$ is the blowup of $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ along $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. This, in turn, is isomorphic to the base change of the blowup $X_{\mathfrak{m}}$ (of $C^{(\deg \mathfrak{m})}$ along $\mathfrak{m}$) via the (flat) projection $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \rightarrow C^{(\deg \mathfrak{m})}$.

Assembling these facts, we obtain the following Cartesian square:

$$\begin{array}{ccccc}
E_{\mathfrak{m}} \times U^{(d-\deg \mathfrak{m})} & \xrightarrow{\tilde{+}} & \overline{\{\eta_0\}} = E_0 & & \\
\downarrow & & \downarrow & & \\
X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{\tilde{+}} & X_{\mathfrak{m},d} & & \\
\downarrow & & \downarrow \pi & & \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{c.i} & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} & \xrightarrow{+} & C^{(d)}
\end{array}$$

The point θ defined in the Corollary is the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Let η be the generic point of $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. By [Corollary 64](#), the map $+$ is étale at η . Consequently, the lifted map $\tilde{+}$ is étale at θ . Given the isomorphism $(+^{-1})(\mathcal{P}^{[d]}) \cong \mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ from [Proposition 28](#), the tame ramification of the box product at θ descends to the tame ramification of $\mathcal{P}^{[d]}$ at η_0 by applying [Lemma 43](#). □

Lemma 68. *Let $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{m}$ be two coprime sub moduli of $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{n}_1]}, \mathcal{P}^{[\deg \mathfrak{n}_2]}$ are at most tamely ramified at $\eta_{\mathfrak{n}_1}, \eta_{\mathfrak{n}_2}$ respectively. Then $\mathcal{P}^{[\deg \mathfrak{n}_1 + \deg \mathfrak{n}_2]}$ is at most tamely ramified at $\eta_{\mathfrak{n}_1 + \mathfrak{n}_2}$.*

Proof. It follows from [Proposition 62](#) and [Proposition 28](#). □

3.3.3 Conclusion of the Tameness Argument

Proof of Theorem 61. Let $\mathcal{P} = \text{Fr}(\mathcal{F})$ be the $\Lambda^\times$ -torsor of frames. For each single-point submodulus $\mathfrak{n} = k_P P$ of $\mathfrak{m}$, [Theorem 3](#) shows that $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$. Repeated application of [Lemma 68](#) shows that $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Therefore [Corollary 67](#) implies that $\mathcal{P}^{[d]}$ is tamely ramified at the generic point η_0 of E_0 for every sufficiently large d . By [Proposition 9](#) and [definition 26](#), the same conclusion holds for $\mathcal{F}^{[d]}$ along H . □

3.4 Extension Across the Boundary

We now use the geometry of the Abel–Jacobi fibers to upgrade the tameness proved above to extension across the boundary. The essential input is that an affine space has no nontrivial finite cover which is tame along the hyperplane at infinity.

Lemma 69. *Let $\mathcal{F}^{[d]}$ be a locally constant sheaf of free rank-one Λ -modules on $U^{(d)}$. If $\mathcal{F}^{[d]}$ is tamely ramified along*

$$H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)},$$

then it extends uniquely to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}_{\mathfrak{m}}^{(d)}$.

Proof. Put $r = d - \deg \mathfrak{m} - g + 1$, let ξ be the generic point of $\text{Pic}_{C,\mathfrak{m}}^d$, and set $\Omega = k(\xi)^{\text{sep}}$. By Takeuchi’s construction, after base change of the generic fiber to $\text{Spec } \Omega$, the compactified Abel–Jacobi fiber and its boundary are

$$\left(\tilde{C}_{\mathfrak{m},\Omega}^{(d)}, H_{\Omega} \right) \cong \left(\mathbb{P}_{\Omega}^r, \mathbb{P}_{\Omega}^{r-1} \right),$$

and the open fiber is $\mathbb{A}_\Omega^r$.

Let $\mathcal{P}^{[d]}$ be the $\Lambda^\times$ -torsor of frames of $\mathcal{F}^{[d]}$. Its restriction to the geometric generic open fiber is tame along the hyperplane at infinity. Tóth proves [Tót11, Example 3.3.8 and Claim 4.2.3] that

$$\pi_1^t(\mathbb{A}_\Omega^r; \mathbb{P}_\Omega^{r-1}) = 1.$$

Let $\bar{\eta}$ be the generic point of the hyperplane $H_\Omega \cong \mathbb{P}_\Omega^{r-1}$. The pullback of $\mathcal{P}^{[d]}$ to $\mathbb{A}_\Omega^r$ is therefore a constant $\Lambda^\times$ -torsor. It extends as the constant finite étale torsor over $\mathbb{P}_\Omega^r$, so the inertia group $I_{\bar{\eta}}$ acts trivially. Let η be the generic point of H . The passage from η to $\bar{\eta}$ gives an unramified extension of the corresponding henselian discrete valuation rings: it enlarges the residue field separably and leaves a uniformizer unchanged. Hence $I_{\bar{\eta}} \cong I_\eta$, and triviality of inertia can be checked after this base change. Thus I_η acts trivially, which says precisely that $\mathcal{P}^{[d]}$ is unramified at the generic point of H .

The scheme $\tilde{C}_m^{(d)}$ is regular, and H is the only codimension-one component of its complement $U^{(d)}$. The standard normalization-and-purity argument, using Zariski–Nagata purity ([Stacks, Tag 0BMB]), therefore extends $\mathcal{P}^{[d]}$ to a finite étale $\Lambda^\times$ -torsor on $\tilde{C}_m^{(d)}$. (Indeed, take the componentwise normalization of $\tilde{C}_m^{(d)}$ in the finite étale cover $\mathcal{P}^{[d]}$. This normalization is finite because $\tilde{C}_m^{(d)}$ is of finite type over a field, hence Nagata. Its branch locus contains no codimension-one point: over $U^{(d)}$ the cover is étale, and at the generic point of H it is unramified by the preceding paragraph. Purity forces every nonempty branch locus over a regular scheme to have a codimension-one component, so the branch locus is empty.) The associated rank-one Λ -module is the desired extension $\tilde{\mathcal{F}}^{[d]}$. It is unique because a dense open immersion into a normal connected scheme induces a surjection on étale fundamental groups. $\square$

Remark. Tameness does not imply unramifiedness for an arbitrary local extension. Here the implication comes from the global fiber geometry: the relevant boundary is the hyperplane at infinity of the geometric generic affine Abel–Jacobi fiber, whose tame fundamental group is trivial. This is the geometric argument used in the tame case by Tóth.

3.5 Descent and Reduced Geometric Class Field Theory

We finish by recording the fundamental-group input and applying it to the ordinary and generalized Abel–Jacobi morphisms.

3.5.1 The Fundamental-Group Input

Proposition 70 ([Stacks, Tag 0C0J]). *Let $f: X \rightarrow S$ be a flat proper morphism of finite presentation whose geometric fibers are connected and reduced. Assume that S is connected, and let $\bar{s}$ be a geometric point of S . Then there is an exact sequence*

$$\pi_1(X_{\bar{s}}) \longrightarrow \pi_1(X) \longrightarrow \pi_1(S) \longrightarrow 1.$$

Corollary 71. *Let $f: X \rightarrow S$ be a proper smooth morphism of finite presentation whose geometric fibers are connected. Then the sequence of Proposition 70 is exact.*

Proof. A smooth geometric fiber is reduced, so this is an immediate application of Proposition 70. $\square$

Corollary 72. *Let $f: X \rightarrow S$ be a projective space bundle over a connected scheme. Then pullback induces an isomorphism*

$$\pi_1^{\text{et}}(X) \xrightarrow{\sim} \pi_1^{\text{et}}(S).$$

Proof. The geometric fibers are projective spaces and therefore have trivial étale fundamental group. The assertion follows from [Corollary 71](#). $\square$

3.5.2 The Descent Argument

We work over $S = \text{Spec } k$, where k is perfect. By [Proposition 9](#), it is equivalent first to prove the following torsor formulation.

Theorem 73 (Torsor formulation). *Let $G = \Lambda^\times$, and let $\mathcal{P}$ be a G -torsor on U whose ramification is bounded by $\mathfrak{m}$. For every sufficiently large integer d , there exists a unique, up to isomorphism, G -torsor $\mathcal{Q}_d$ on $\text{Pic}_{C,\mathfrak{m}}^d$ such that*

$$\Phi_d^* \mathcal{Q}_d \cong \mathcal{P}^{[d]}.$$

Proof. We divide the proof into two cases.

Case 1: $\mathfrak{m} = 0$. For d sufficiently large, [Theorem 59](#) identifies

$$\Phi_d: C^{(d)} \longrightarrow \text{Pic}_C^d$$

as a projective space bundle. Hence [Corollary 72](#) gives

$$\pi_1^{\text{et}}(C^{(d)}) \xrightarrow{\sim} \pi_1^{\text{et}}(\text{Pic}_C^d).$$

The G -torsor $\mathcal{P}^{[d]}$ therefore descends uniquely to a G -torsor on Pic_C^d .

Case 2: $\mathfrak{m} > 0$. Let $\mathcal{F} = \mathcal{P} \times^G \Lambda$ be the rank-one local system associated with $\mathcal{P}$. By [Definition 26](#), the frame torsor of $\mathcal{F}^{[d]}$ is $\mathcal{P}^{[d]}$. For d sufficiently large, [Theorem 60](#) extends Φ_d to a projective space bundle

$$\tilde{\Phi}_d: \tilde{C}_{\mathfrak{m}}^{(d)} \longrightarrow \text{Pic}_{C,\mathfrak{m}}^d.$$

By [Theorem 61](#), $\mathcal{F}^{[d]}$ is tamely ramified along $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$. Hence [Lemma 69](#) gives a unique extension $\tilde{\mathcal{F}}^{[d]}$ to $\tilde{C}_{\mathfrak{m}}^{(d)}$.

By [Corollary 72](#),

$$\pi_1^{\text{et}}(\tilde{C}_{\mathfrak{m}}^{(d)}) \xrightarrow{\sim} \pi_1^{\text{et}}(\text{Pic}_{C,\mathfrak{m}}^d).$$

Thus $\tilde{\mathcal{F}}^{[d]}$ descends uniquely to a rank-one local system $\mathcal{G}_d$ on $\text{Pic}_{C,\mathfrak{m}}^d$. Restricting to $U^{(d)}$ gives $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$. Taking frame torsors yields

$$\Phi_d^* \text{Fr}(\mathcal{G}_d) \cong \mathcal{P}^{[d]},$$

so $\mathcal{Q}_d = \text{Fr}(\mathcal{G}_d)$ has the required property. Uniqueness follows from the fundamental-group isomorphism and from the surjection

$$\pi_1^{\text{et}}(U^{(d)}) \twoheadrightarrow \pi_1^{\text{et}}(\tilde{C}_{\mathfrak{m}}^{(d)})$$

induced by the dense open immersion into the normal scheme $\tilde{C}_{\mathfrak{m}}^{(d)}$. $\square$

Proof of Theorem 2. Let $\mathcal{F}$ be as in Theorem 2, and set $\mathcal{P} = \text{Fr}(\mathcal{F})$. Apply Theorem 73 to obtain $\mathcal{Q}_d$ on $\text{Pic}_{C, \mathfrak{m}}^d$, and let

$$\mathcal{G}_d = \mathcal{Q}_d \times^{\Lambda^\times} \Lambda.$$

The equivalence of Proposition 9, together with the compatibility of frame torsors and symmetric powers from Definition 26, gives $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$. The same equivalence carries uniqueness of $\mathcal{Q}_d$ to uniqueness of $\mathcal{G}_d$. $\square$